



Blinkit Business Dashboard

1. Project Overview

You are working as a **Data Analyst at Blinkit**, one of India's leading **quick-commerce** companies.

Your task is to **analyze product performance, outlet operations, sales trends, and customer satisfaction** using **Power BI**, and present insights to management through an **interactive dashboard**.

Business Problem

Blinkit's management has identified key business challenges:

- Certain product categories are underperforming despite high demand.
- Outlet performance varies significantly across **Tier 1, Tier 2, and Tier 3** cities.
- Sales are affected by **item visibility** and **fat content labeling**.
- **Customer satisfaction** and **repeat purchases** require close monitoring.

2 Dataset Provided

Blinkit Grocery Dataset → [Click here to get Dataset](#)

Category	Columns
Item Details	Item Identifier, Item Type, Item Fat Content, Item Weight, Item Visibility

Sales Details	Sales, Rating, Outlet Identifier
Outlet Details	Outlet Establishment Year, Outlet Location Type (Tier 1/2/3), Outlet Size, Outlet Type

3 Data Cleaning & Transformation

Perform all necessary **data cleaning and transformation** steps using **Power Query** to prepare the dataset for modeling and analysis.

Guidelines

- Ensure columns are well-formatted and consistent.
- Handle missing or incorrect data appropriately.
- Maintain correct data types for numerical and categorical fields.
- Remove duplicate or irrelevant records.
- Create relevant calculated columns such as:
 - *Sales per Kg*
 - *Years of Operation*
 - *Profit Margin* (if cost data is available)
- Prepare lookup tables for:
 - Item details
 - Item nutrition (fat content)
 - Outlet information
 - Outlet location

4 Data Modeling

Design a **relational star schema model** connecting fact and dimension tables.

Suggested Structure

- **Fact Table:** Sales
- **Dimension Tables:** Items, Item Content, Outlets, Outlet Location
- **Optional Table:** Calendar (if order date is available)

Relationships:

Many-to-one relationships between the **fact table** and corresponding **dimension tables**.

5 DAX Measures

Create **DAX measures** to answer key business questions and KPIs.

Metric	Description
Total Sales	Total revenue generated across all outlets
Average Sales per Outlet	Average outlet performance
Top Performing Items	Items with highest sales
Profit Margin %	Percentage of profit retained (if cost available)
Average Item Rating (CSAT)	Overall customer satisfaction

Sales by Fat Content	Comparison of Regular vs Low Fat items
Sales by Tier	Performance across Tier 1/2/3 cities
Sales by Outlet Type	Insights by outlet format
Sales Trend	Month-over-month or year-over-year growth
Item Visibility Impact	Correlation between visibility and sales

6 Dashboard Design

Your dashboard should provide a **clear, interactive, and insight-driven view** of the business.

A. KPI Cards

- Total Sales
- Average Sales per Outlet
- Average Item Rating (CSAT)
- Profit Margin % (if applicable)
- Sales per Kg
- Average Years of Operation

B. Visuals (Recommended)

- **Bar / Column Charts:** Sales by Item Type, Outlet Type, and Tier
- **Line Chart:** Monthly / Yearly Sales Trend
- **Pie / Donut Chart:** Fat Content Distribution

- **Funnel Chart:** Sales Conversion by Outlet Type
- **Gauge Chart:** Sales vs Target
- **Scatter Chart:** Item Visibility vs Sales
- **Matrix / Table:** Top 10 Items by Sales and Rating

C. Interactivity & Formatting

- Add **Slicers** for Tier, Outlet Type, Item Type, and Fat Content.
- Enable **Drill-through navigation:** Tier → Outlet Type → Outlet → Item.
- Apply **Conditional Formatting:** Highlight outlets with below-average sales.
- Ensure all visuals are **interactive and responsive**.

7 Business Questions to Answer

Use your dashboard to answer the following management questions:

1. Which product categories contribute most to overall sales?
2. How does fat content (Regular vs Low Fat) influence sales?
3. Which outlets (by size, type, or tier) perform best?
4. What is the trend of sales growth across establishment years?
5. Which items have high visibility but low sales (underperforming products)?
6. Who are the Top 10 items by sales and rating?
7. How does customer satisfaction vary across categories, outlet types, and tiers?

8 Deliverables

Students must submit the following:

1. **Cleaned & Transformed Data Model** in Power BI
2. **DAX Measures** created for defined KPIs
3. **Interactive Dashboard** with required visuals and slicers
4. **Business Insights Report (1–2 pages)** summarizing recommendations and findings